Aaryan S. Maralihalli

Machine Learning Engineer

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Summary

ML engineer who takes models from research to deployment — DL on signals, **LLM fine-tuning (LoRA)**, and serving via FastAPI. **IEEE-published**: built a self-supervised cardiac model (**ROC-AUC 0.757**); fine-tuned Mistral-7B for sub-100ms retrieval. Healthcare-ML domain depth. CS graduate, Jul 2026.

Experience

HyperVerge, Bangalore
Integration Engineer Intern Jan 2026 – Jun 2026
AI identity-verification / KYC platform

- Integrated and optimized **production AI/ML verification modules** (face match, liveness, name match) for 6 enterprise fintech clients; drove model-scoring changes and post-go-live performance tuning (conversion/drop-off).

Integrated Genetic Solutions (IGS), Bangalore
Project Intern, Medical AI Jun – Aug 2024

- Worked on ML workflows for medical document processing on a HIPAA-compliant lab-value extraction product.

Selected Projects

Self-Supervised Cardiac Risk Model (capstone / published)github.com/Aaryansm11/ecg-ppg-arrhythmia-prediction

- **Contrastive learning (SimCLR / NT-Xent)** on **MIMIC-III** ECG/PPG — 133K segments; custom 1D-CNN encoder (128-d embeddings) + Random Forest clinical model; **ROC-AUC 0.757**; **FastAPI** inference server + XAI (saliency, integrated gradients).

LANGTRON github.com/Aaryansm11/LANGTRON

- **Fine-tuned Mistral-7B (LoRA)** with a dual-model architecture; **FAISS** semantic retrieval at **sub-100ms**.

FinFusion

- Built the **core multimodal forecasting model** (LSTM time-series + transformer news-sentiment fusion), improving MAPE 15–20% over single-modal baselines.

Langtron-KGS github.com/Aaryansm11/Langtron-KGS

- **SpaCy NER + BERT relation extraction** → Neo4j knowledge graph (FastAPI).

Skills

- **ML/DL:** PyTorch, HuggingFace Transformers, contrastive/self-supervised learning, LoRA fine-tuning, CNNs, Random Forest, XAI
- **LLM/NLP:** LangChain, FAISS, spaCy, BERT, RAG, embeddings
- **Serving/Infra:** FastAPI model serving, Docker, AWS, Python, SQL; DSA (150+ LeetCode)

Publication & Education

- **IEEE ICASI 2026 (Kyoto), Oral Presentation** — *"An Interpretable Model for Stroke Prediction and Arrhythmia Classification using ECG & PPG."*
- **B.Tech, Computer Science** — PES University, Bangalore — 2022 – Jul 2026